A. BrainImation Multimodal Client-Side Architecture: Muse S Athena + Classic BLE Client, PPG/fNIRS, 6-DOF IMU & Visual Catalogue

1. Multi-Gen BLE Client (muse-browser.js, 37.4 KB)

- Classic Two-Phase Lazy GATT (Muse 2016 / Muse 2 / Muse S)
- Muse S Athena (273e0013): 14-bit EEG, 20-bit 16-ch PPG, 52 Hz IMU, DRL/REF, Battery
- Live p1042 / p50 PPG Toggle

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2. Multimodal Signal Engine (window.eegData + Sim)

- 256 Hz EEG + 60/50Hz Notch + FFT
- 64 Hz PPG, HR, SpO_2, 730/850nm MBLL $\Delta[HbO_2]/\Delta[HHb]$ + 8/20mm Short-Channel Skin Regression
- 52 Hz 6-DOF IMU Kinematics + 6-Tap NLMS Motion Filter

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3. Browser IDE & Catalogue (Monaco + 153-API Ref)

- Resizable Visual Catalogue (234 Cards, 21 Categories)
- 5-Tab Reference Drawer (+ AI Tab): P5, Input, Sound, EEG, IMU/PPG
- In-Canvas About Guide & Zoom-Style Mobile Fullscreen

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4. Dynamic Introspection (bindP5Functions + BGShared)

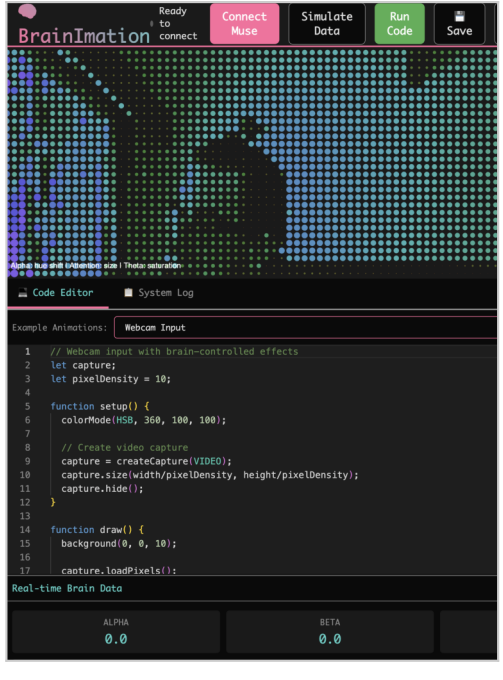
- Auto-binds p5.js 2D/WebGL & p5.sound to window
- Forwards 15 Lifecycle/DOM hooks
- BGShared Arcade CRT Engine
- CSV Exporter & SoundTester

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5. Live Output Modalities (234 Catalogue Sketches)

- 44 Starters, Diag & Core
- 26 PPG/fNIRS, IMU & Multi
- 14 Closed-Loop BrainGames
- 29 EEGEdu & p5.eegedu Ports
- 121 Student Archive Sketches

B. BrainImation Live Browser IDE & Visual Catalogue



C. Unified 21-Category Visual Catalogue (234 Interactive Sketches with Thumbnail Cards)

Catalogue Category / Track	N	Med.	Representative Sketches	Primary Physiological & API Modalities
1–2. One-Sensor Starters & Diag.	7	25	simpleEEG, simplePPG, athenaPresetSweep	Minimal starters & Athena 16-ch diagnostics
3–5. Intro, Bands & Raw EEG	13	32	basic, bandOrbitz, epochs, SpO2Fused	eegData bands, 256 Hz ERPs, fused SpO_2
6. PPG & Optical (fNIRS Studio)	10	200	fNIRSBeerLambert, ppgShortChannel...	64 Hz 660/730/850nm, HbO_2/HHb, 8/20mm OLS
7. Motion, IMU & Broadcast	11	215	imuStackedKinematics, imuHeadMove...	52 Hz Accel/Gyro, 2nd-order pos., NLMS filter
8. Multimodal (EEG + PPG + HR)	5	117	multiNeurovascularCoupling, HEP	Neurovascular lag, PPG-peak locked HEP
9–15. Attention, 3D & Sound	24	34	mandala, 3dTorus, perlinNoise, brainSynth	WEBGL 3D, particles, webcam, Chowning FM
16. BrainGames Arcade Suite	14	514	gameGolfShooter, gameArchery, FPS	14 closed-loop BCI arcade games (BGShared)
17–18. EEGEdu & p5.eegedu	29	44	eegeduFlock3D, Oddball, eegeduEpochs	Per-channel PSD, SSVEP/ERP CSV export
19–20. 2025 Neurotech Class	95	149	EegFlappyBird, EegTetris, CatErp	Curated PSYCH 403 BCI games & 4-ch ERPs
21. 2022 NeuroArt Class Gallery	26	43	studentBrainTyphoon, Terrain3D	Inaugural 2022 undergraduate neuro-art gallery